

A Lightweight Hybrid Machine Learning Framework for Phishing URL Detection and Classification

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Abstract:

Background/Introduction

The rapid proliferation of digital services is consistently accompanied by sophisticated phishing attacks. Traditional static blacklisting struggles against dynamically generated zero-hour malicious URLs. Consequently, this research will leverage machine learning as a proactive approach to identify phishing threats by analyzing structural and lexical URL characteristics without the high computational costs of deep neural networks.

Objectives

The primary objective will be to design and evaluate a fast, robust, and lightweight hybrid machine learning framework for phishing classification. This study aims to achieve exceptional accuracy while remaining computationally efficient for real-time deployment. Additionally, the research will validate the framework's generalizability across diverse datasets and interpret its decision boundaries using explainable AI.

Methods/Methodology

The proposed methodology will involve preprocessing the comprehensive PhiUSIIL Phishing URL Dataset alongside a secondary validation dataset. The pipeline will implement strict null imputation and standard scaling. We will train and compare Logistic Regression, Random Forest, and XGBoost against a novel soft-voting hybrid ensemble. Cross-validation will ensure unbiased metrics, while SHAP will extract actionable feature importance.

Results/Findings

It is anticipated that XGBoost will emerge as the top standalone performer in raw accuracy and inference latency. However, the soft-voting hybrid classifier is expected to demonstrate superior stability during cross-dataset generalization. We project that SHAP analysis will isolate critical malicious markers, such as excessive external references, decoding the ensemble's decision boundaries.

Conclusion

This study will demonstrate that carefully engineered, lightweight hybrid frameworks can effectively neutralize zero-hour phishing URLs. By prioritizing computational efficiency and resilience over model complexity, the resulting framework will provide a highly viable foundation for future integration into real-time, edge-deployed cybersecurity applications.

Keywords: Machine Learning, Phishing Detection, XGBoost, Ensemble Learning, Cyber Threats